组会报告

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Motivation

Li, X., et al. (2020) Semantic Flow for Fast and Accurate Scene Parsing. arXiv e-prints arXiv:2002.10120

影响分割性能的两个关键因素：

1.详细的分辨率信息。

2.强大的语义表征。

快速模型在性能上没有很大的优势。主要原因在于深层到浅层的不高效的语义传递。

解决方法：受光流的启发，在不同层不同分辨率下学习语义流。

Method

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Flow Alignment Module

FPN 的设计看起来是对称的编码和解码器，但是双线性上采样打破了这种对称性。

双线性上采样均匀插值一个特征图，但是特征图之间的非对齐很复杂。因此需要显式建立特征图的一致性。

类似于通过光流对齐两个视频帧，通过预测一个流场，来对齐两个近邻层级的特征图。

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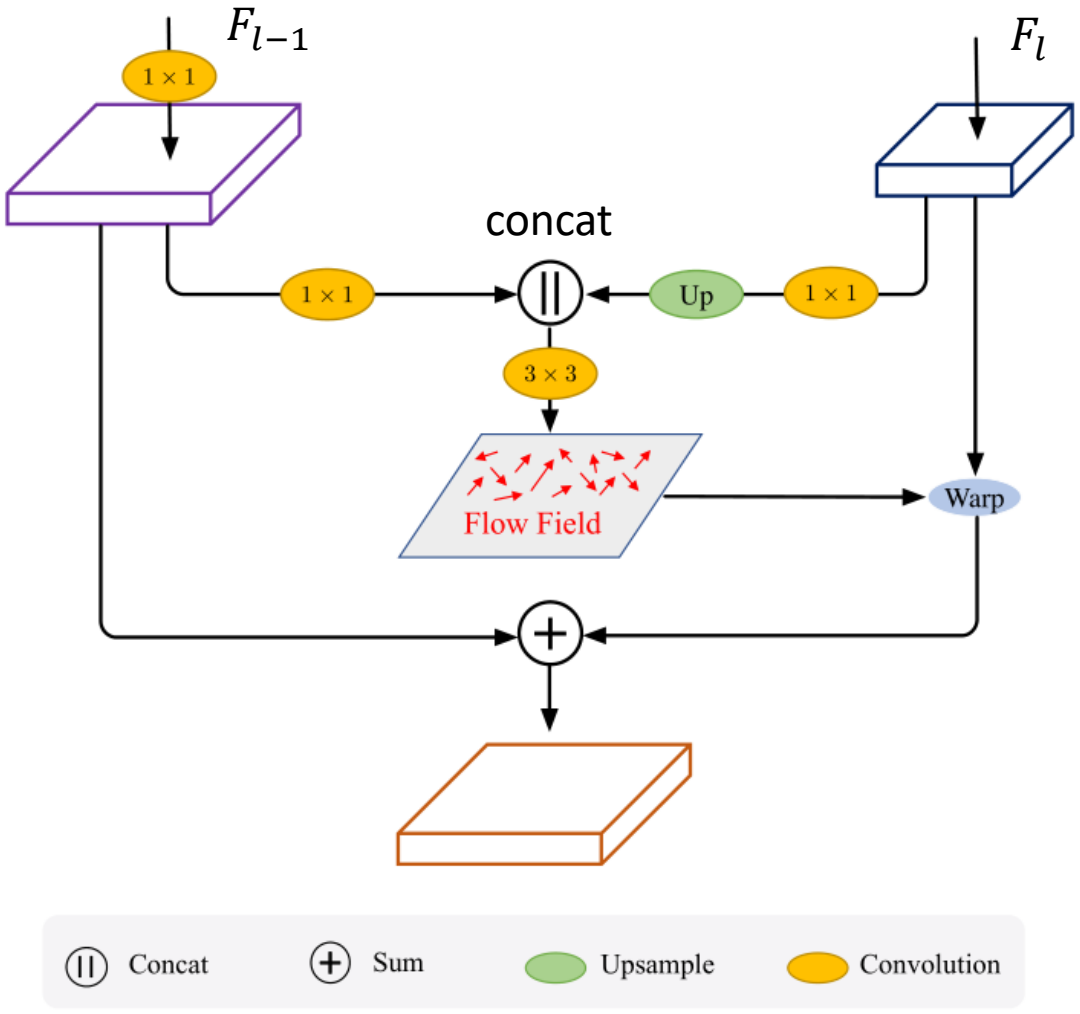

(a) Flow Alignment Module

给定两个近邻特征图 F_l 和 F_{l-1}，首先将 F_l 通过双线性插值到和 F_{l-1} 一样大小，然后把它们级联，通过卷积预测语义流场 $\Delta_{l-1} \in R^{H_{l-1} \times W_{l-1} \times 2}$ （偏置）。

每个像素 p_{l-1} 都通过 $\frac{p_{l-1}+\Delta_{l-1}(p_{l-1})}{2}$ 映射到像素 p_l。

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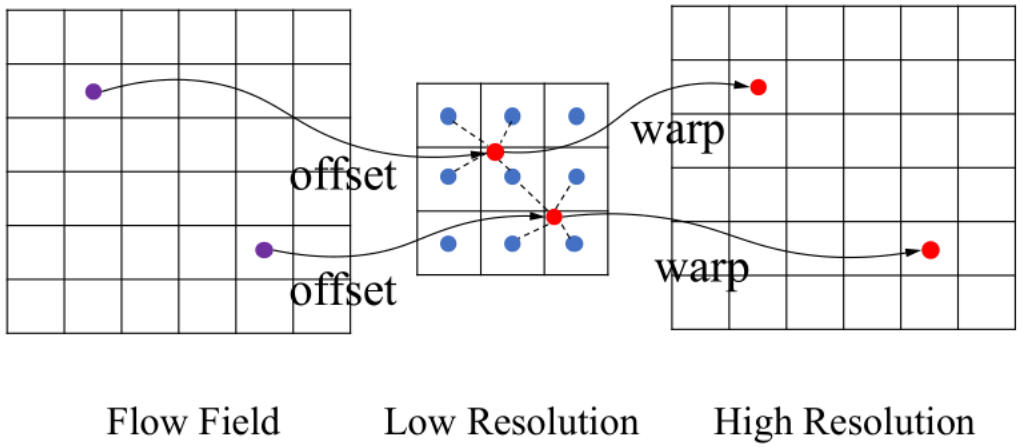

(b) Warp Procedure

然后使用双线性采样，线性插值 p_l 的四个近邻像素来近似 $\tilde{F}_l(p_{l-1})$。

$$\tilde{F}_l(p_{l-1}) = F_l(p_l) = \Sigma_{p \in \mathcal{N}(p_l)} \omega_p F_l(p)$$

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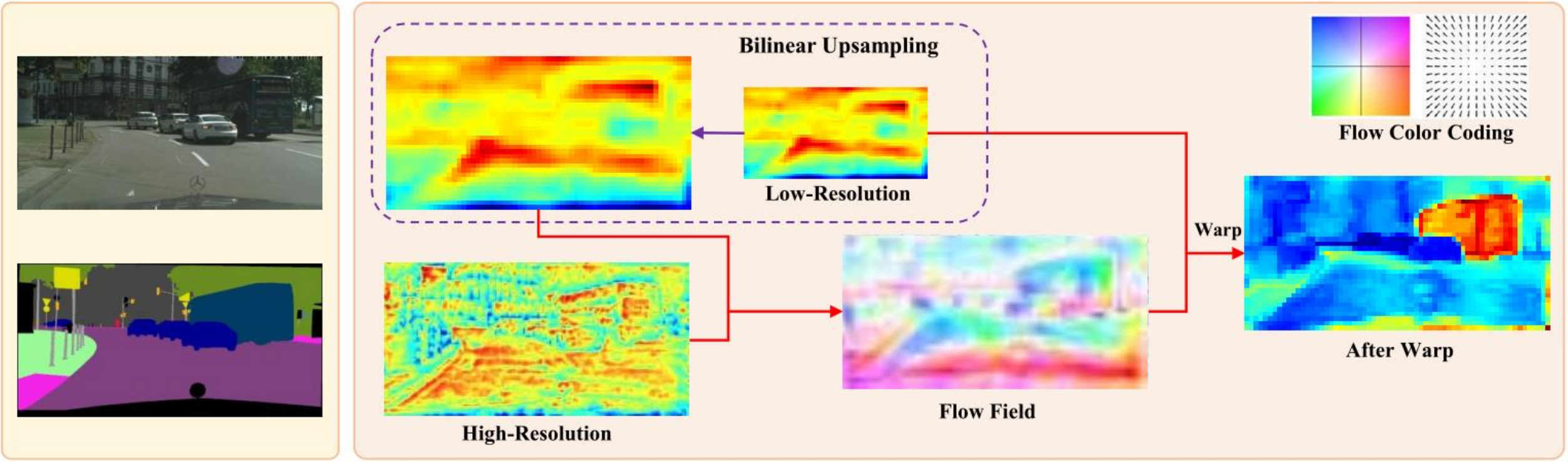

Figure 2. Visualization of feature maps and semantic flow field in FAM. Feature maps are visualized by averaging along the channel dimension, where large values are denoted by hot colors and vice versa. For visualizing semantic flow field, color code proposed by [2] and showed in top-right corner is adopted, where orientation and magnitude of flow vectors are represented by hue and saturation respectively.

Architecture

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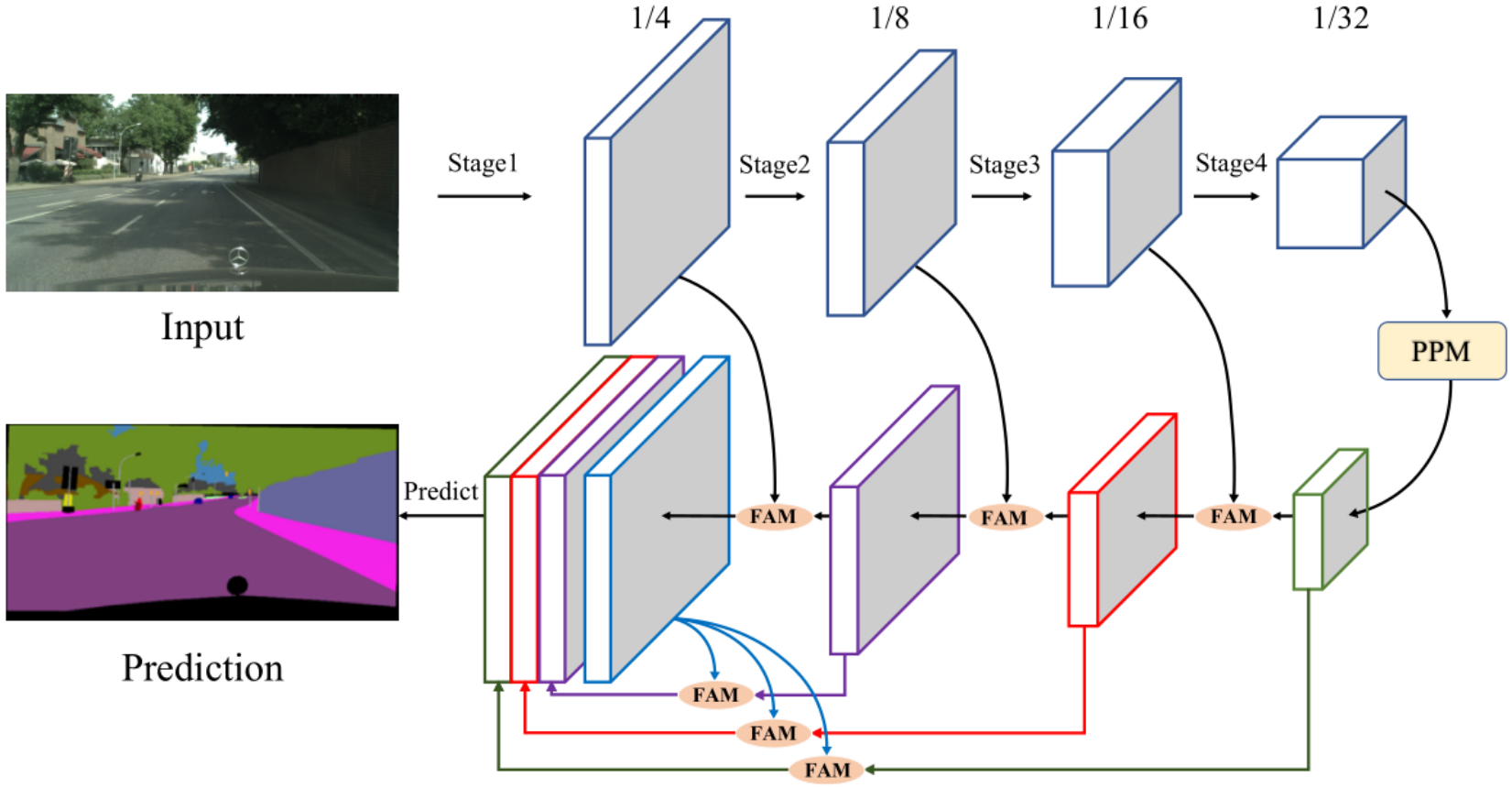

FAM: Feature alignment module
PPM: pyramid pooling module

Experiments

Yu, C., et al. (2020). Context Prior for Scene Segmentation. CVPR 2020.

Method	Stride	mIoU (%)	Δa
FCN	32	71.5	-
Dilated FCN	8	72.6	1.1 ↑
+FPN	32	74.8	3.3 ↑
+FAM	32	77.2	5.7 ↑
+FPN + PPM	32	76.6	5.1 ↑
+FAM + PPM	32	**78.7**	7.2 ↑

Table 1. Ablation study on baseline methods, where ResNet-18 serves as the backbone.

Method	$\mathbf{F}_3$	$\mathbf{F}_4$	$\mathbf{F}_5$	mIoU(%)	Δa
FPN+PPM				76.6	-
	✓			76.9	0.3 ↑
		✓		77.0	0.4 ↑
			✓	77.5	0.9 ↑
		✓	✓	77.8	1.2 ↑
	✓	✓	✓	78.3	1.7 ↑

Table 2. Ablation study on different positions to insert FAM, where $\mathbf{F}_l$ denote the upsampling position between level l and level $l-1$, ResNet-18 with FPN decoder and PPM head serves as baseline.

Experiments

Yu, C., et al. (2020). Context Prior for Scene Segmentation. CVPR 2020.

Method	mIoU(%)	Δa	#GFLOPs
FAM	76.4	-	-
+PPM [60]	78.3	1.9↑	246.5
+NL [46]	76.8	0.4↑	295.5
+ASPP [5]	77.6	1.2↑	276.6
+DenseASPP [49]	77.5	1.1↑	282.3

Table 3. Ablation with different contextual modeling heads, where ResNet-18 with FAM serves as the baseline.

Backbone	mIoU(%)	Δa	#GFLOPs
ResNet-50 [17]	76.8	-	664.7
w/ FAM	79.2	2.4 ↑	673.1
ResNet-101 [17]	77.6	-	824.9
w/ FAM	79.8	2.2↑	833.5
ShuffleNetv2 [30]	69.8	-	35.3
w/ FAM	72.1	2.3 ↑	35.9
DF1 [48]	72.1	-	36.8
w/ FAM	74.3	2.2 ↑	36.9
DF2 [48]	73.2	-	96.0
w/ FAM	75.8	2.6 ↑	96.5

Table 4. Ablation study on different backbones, where FPN decoder with PPM head is used as baseline. The top part compare deep networks with sliding window testing, and the bottom part compares light-weight networks using single view testing.

Experiments

Yu, C., et al. (2020). Context Prior for Scene Segmentation. CVPR 2020.

Method	InputSize	mIoU (%)	#FPS	#Params
ENet [39]	640 × 360	58.3	60	0.4M
ESPNet [32]	512 × 1024	60.3	132	0.4M
ESPNetv2 [33]	512 × 1024	62.1	80	0.8M
ERFNet [41]	512 × 1024	69.7	41.9	-
BiSeNet(ResNet-18) [50]	768 × 1536	74.6	43	12.9M
BiSeNet(Xception-39) [50]	768 × 1536	68.4	72	5.8M
ICNet [59]	1024 × 2048	69.5	34	26.5M
DF1-Seg [48]	1024 × 2048	73.0	80	8.55M
DF2-Seg [48]	1024 × 2048	74.8	55	8.55M
SwiftNet [38]	1024 × 2048	75.5	39.9	11.80M
SwiftNet-ens [38]	1024 × 2048	76.5	18.4	24.7M
DFANet [23]	1024 × 1024	71.3	100	7.8M
CellNet [58]	768 × 1536	70.5	108	-
SFNet(DF1)	1024 × 2048	**74.5**	74/121	9.03M
SFNet(DF2)	1024 × 2048	**77.8**	53/61	10.53M
SFNet(ResNet-18)	1024 × 2048	**78.9**	18/26	13.47M
SFNet(ResNet-18)†	1024 × 2048	**80.4**	18/26	13.47M

†Mapillary dataset used for pretraining.

Table 6. Comparison on Cityscapes *test* set with state-of-the-art real-time models. For fair comparison, input size is also considered, and all models use single scale inference.

Experiments

Yu, C., et al. (2020). Context Prior for Scene Segmentation. CVPR 2020.

Method	Backbone	mIoU (%)	#Params	#GFLOPs†
SAC [57]	ResNet-101	78.1	-	-
DepthSeg [22]	ResNet-101	78.2	-	-
PSPNet [60]	ResNet-101	78.4	65.7M	2128.2
BiSeNet [50]	ResNe-18	77.7	12.3M	
BiSeNet [50]	ResNet-101	78.9	51.0M	437.4
DFN [51]	ResNet-101	79.3	90.7M	2241.3
PSANet [61]	ResNet-101	80.1	85.6M	2364.4
DenseASPP [49]	DenseNet-161	80.6	35.7M	1265.2
SPGNet [7]	2×ResNet-50	81.1	-	-
ANNet [66]	ResNet-101	81.3	63.0M	2178.7
CCNet [18]	ResNet-101	81.4	66.5M	2307.0
DANet [13]	ResNet-101	81.5	66.6M	2596.8
SFNet	ResNet-18	**79.5**	13.47M	246.5
SFNet	ResNet-101	**81.3**	50.32M	833.5

† #GFLOPs calculation adopts 1024×1024 image as input.

Table 7. Comparison on Cityscapes *test* set with state-of-the-art accurate models. For better accuracy, all models use multi-scale inference.

Experiments

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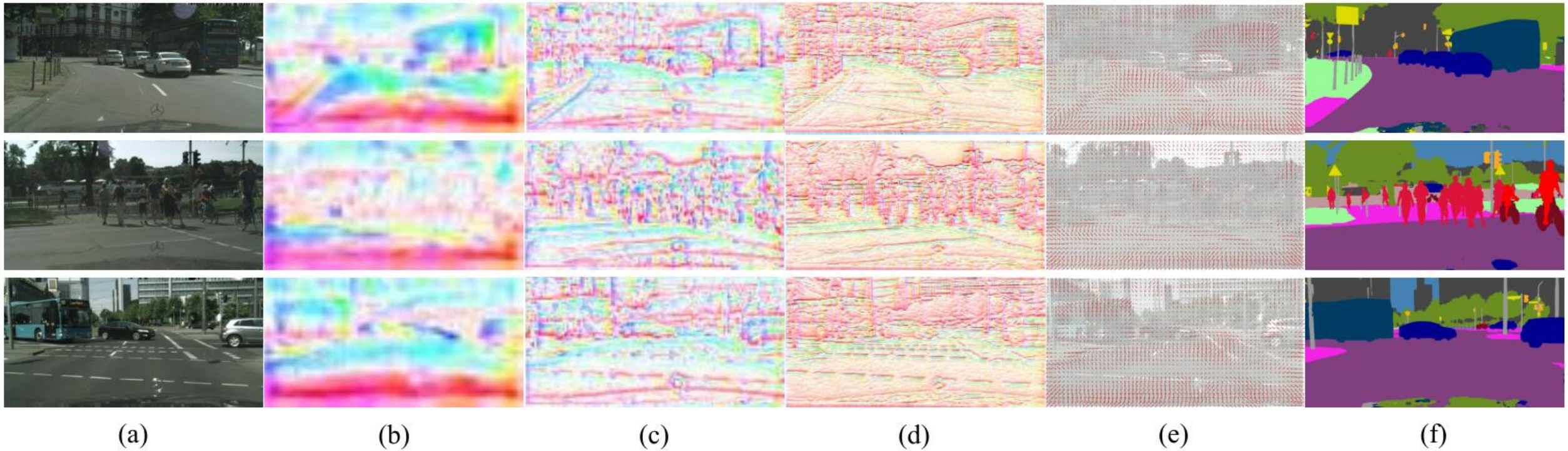

Figure 5. Visualization of the learned semantic flow fields. Column (a) lists three exemplar images. Column (b)-(d) show the semantic flow of the three FAMs in an ascending order of resolution during the decoding process, following the same color coding of Figure 2. Column (e) is the arrowhead visualization of flow fields in column (d). Column (f) contains the segmentation results.

Experiments

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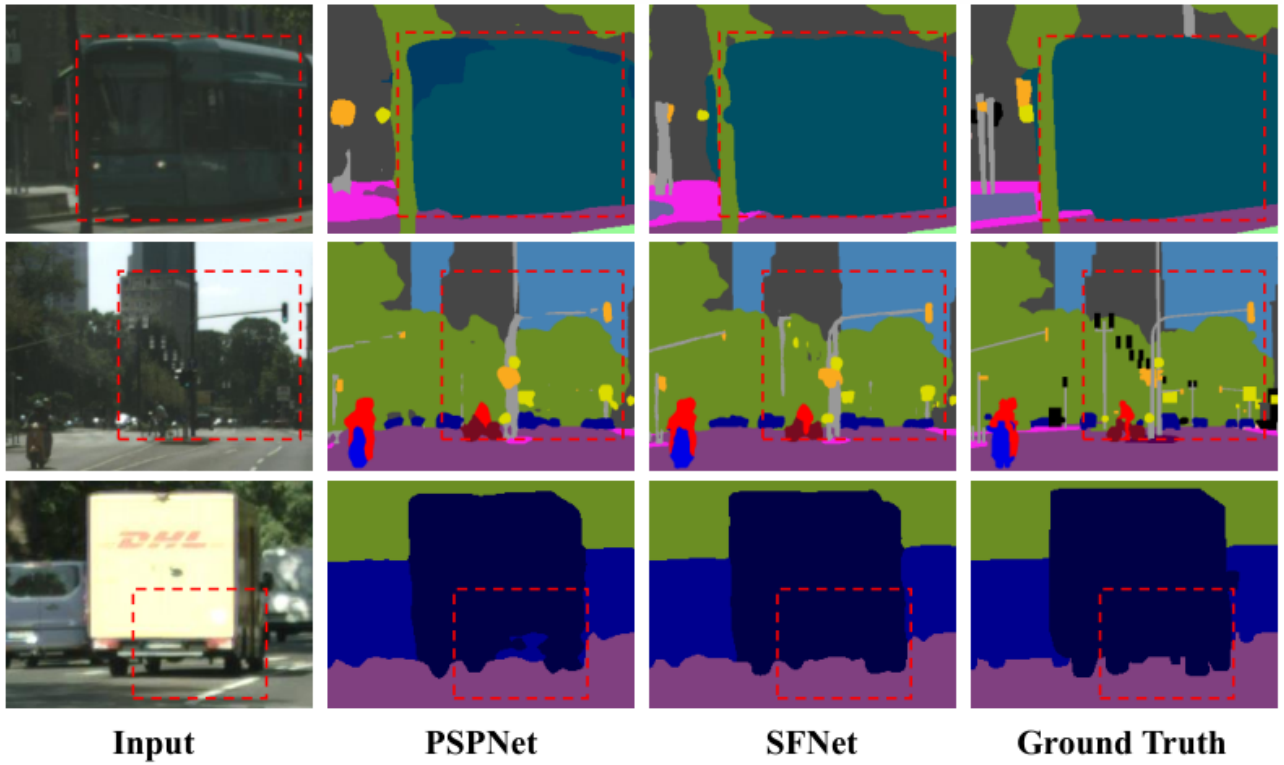

Figure 7. Scene parsing results comparison against PSPNet [60], where significantly improved regions are marked with red dashed boxes. Our method performs better on both small scale and large scale objects.